

CHAPTER 6

FIXED POINTS

The primary significance of the Recursion Theorem **5.6** is foundational, since the result justifies on the basis of the axioms a method of definition of functions which is intuitively obvious. From a purely mathematical point of view, however, we can also view **5.6** as a theorem of *existence and uniqueness* of solutions for systems of identities of the form

$$\begin{aligned} f(0) &= a, \\ f(Sx) &= h(f(x)) \quad (x \in \mathbb{N}), \end{aligned} \tag{6-1}$$

where $a \in E$ and $h : E \rightarrow E$ are given and the function $f : \mathbb{N} \rightarrow E$ is *the unknown*. In this chapter we will prove an elegant generalization of the Recursion Theorem in the context of the theory of partial orderings, which implies the existence and uniqueness of solutions for systems of functional identities much more general than (6-1). The CONTINUOUS LEAST FIXED POINT THEOREM **6.21** is fundamental for the *theory of computation*, it is the basic mathematical tool of the so-called *fixpoint theory of programs*. In the next chapter we will show that it is a special case of a much deeper FIXED POINT THEOREM of Zermelo, which is intimately related to the theory of wellorderings and rich in set theoretic consequences: for example it implies directly the Hypothesis of Cardinal Comparability, **3.1**. Thus, in addition to its purely mathematical significance, the Continuous Least Fixed Point Theorem yields also an interesting point of contact between classical set theory and today's theoretical computer science.

In their simplest and most natural expressions, the Fixed Point Theorems are somewhat abstract and apparently unrelated to the solution of systems of functional identities to which we intend to apply them. To understand what they say and how to use them, we will need to introduce first some basic notions from the theory of partial orderings.

6.1. A partially ordered set or simply **poset** is a structured set

$$P = (\text{Field}(P), \leq_P),$$

where $\text{Field}(P)$ is an arbitrary set and $\leq_P$ is a partial ordering on $\text{Field}(P)$, i.e., a reflexive, transitive and antisymmetric binary relation. Notice that $\leq_P$ determines P because it is reflexive,

$$x \in \text{Field}(P) \iff x \leq_P x,$$



FIGURE 6.1. A discrete and a flat poset.

so we can specify a poset P completely by defining its partial ordering $\leq_P$. In practice, however, the partial ordering $\leq_P$ is often clear from the context and we will tend to identify a poset P with its field $\text{Field}(P)$, following the general convention about structured sets discussed in 4.30. For example, by *the poset* $\mathbb{N}$ we obviously mean the pair $(\mathbb{N}, \leq)$, where $\leq$ is the usual ordering on the set of natural numbers. By this convention, *the points of* P are the members of $\text{Field}(P)$, a *subset* $I \subseteq P$ is a subset $I \subseteq \text{Field}(P)$, etc. Each $I \subseteq P$ is a poset in its own right, partially ordered by the restriction of $\leq_P$ to I ,

$$x \leq_I y \iff_{\text{df}} x \leq_P y \ \& \ x \in I \ \& \ y \in I, \quad (6-2)$$

which is (easily) a partial ordering. We will often deal with posets which have a least element, and it will be convenient to use the same, standard symbol $\perp$ (read “bottom” or “least”) for it, just as we use the same symbol 0 for the additive unit of every number system:

$$\perp = \perp_P =_{\text{df}} \text{the least element of } P \text{ (if it exists)}. \quad (6-3)$$

Any set A can be viewed as a **discrete poset** in which no two elements are comparable, i.e., partially ordered by the identity relation

$$x \leq y \iff x = y \quad (x, y \in A).$$

Just above these in complexity are the **flat posets** which have a least element, the only element involved in any comparisons: i.e.,

$$x \leq_P y \iff x = \perp \vee x = y.$$

The simplest non-empty poset is a singleton $\{\perp\}$, which is both discrete and flat. Additional examples of posets are the sets of natural numbers $\mathbb{N}$, rationals $\mathbb{Q}$ and reals $\mathbb{R}$, with their usual orderings. These are all linear (totally ordered) posets. There is a large variety of finite posets and their study constitutes an important research area of mathematics, but we will not be much concerned with it here. Mostly we will use them as counterexamples. In drawing posets we indicate $x < y$ by placing y above or to the right of x and drawing a polygonal line (sometimes directed, to avoid ambiguity) from x to y , which may pass through other points, e.g., $c < e$ in Figure 6.2.¹³

6.2. Definition. Let P be a poset, $S \subseteq P$ and $M \in P$ a member of P .

¹³There are those who draw posets growing *to the right*, those who draw them growing *up* and even those who draw them growing *down*; to the best of my knowledge, nobody pictures posets growing to the left and it does not appear that any of the three common choices is dominant.

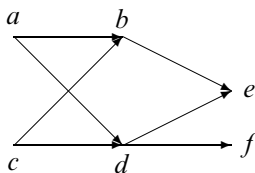


FIGURE 6.2. A finite poset

1. M is an **upper bound** of S if it is greater than or equal to every element of S , i.e., if $(\forall x \in S)[x \leq M]$.
2. M is **maximum** in S if it is a member and an upper bound of S , i.e., if $M \in S$ & $(\forall x \in S)[x \leq M]$.
3. M is a **least upper bound** of S if it is an upper bound and also less than or equal to every other upper bound of S , i.e., if

$$(\forall x \in S)[x \leq M] \& (\forall M')[(\forall x \in S)[x \leq M'] \implies M \leq M'].$$

If M_1, M_2 are both least upper bounds of S , then $M_1 \leq M_2$ (because M_2 is an upper bound and M_1 is a least upper bound) and symmetrically $M_2 \leq M_1$, so that $M_1 = M_2$, i.e., S has at most one least upper bound. When it exists, the least upper bound of a set S is denoted by

$$\sup S = \text{the least upper bound of } S. \quad (6-4)$$

The term “sup” from the Latin *supremum* (maximum) is justified by the following observation.

6.3. Exercise. If M is maximum of a set S in a poset P , then M is also the least upper bound of S .

6.4. Exercise. In the poset of Figure 6.2, find subsets S with the following properties: (1) S has no upper bound. (2) S has upper bounds but no least upper bound. (3) S has a least upper bound but no maximum element.

6.5. Exercise. In any poset P , an element M is the least upper bound of the empty set $\emptyset$ if and only if M is the least element of P , i.e.,

$$\perp = \sup \emptyset \quad (6-5)$$

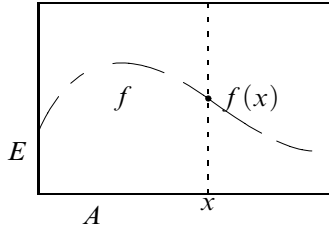
if $\perp$ or $\sup \emptyset$ exists.

6.6. Exercise. The powerset $\mathcal{P}(A)$ of every set A is partially ordered by the relation

$$X \subseteq_A Y \iff_{\text{df}} X \subseteq Y \subseteq A,$$

so that $\perp = \emptyset$ and for every $S \subseteq \mathcal{P}(A)$, the union $\bigcup S$ is the least upper bound of S .¹⁴

¹⁴The partial ordering of $\mathcal{P}(A)$ is the restriction $\subseteq_A$ of the definite condition $X \subseteq Y$ to $\mathcal{P}(A)$, and we often skip the subscript in referring to it, cf. 4.8.

FIGURE 6.3. A partial function $f : A \rightarrow E$.

Less trivial and more interesting for our purposes is the next example of a poset.

6.7. Definition. A **partial function** on a set A to a set E is any function with domain of definition some subset of A and values in E , in symbols

$$f : A \rightarrow E \iff_{\text{df}} \text{Function}(f) \ \& \ \text{Domain}(f) \subseteq A \ \& \ \text{Image}(f) \subseteq E. \quad (6-6)$$

For example, $(n \mapsto (n - 1))$ is a partial function on the natural numbers defined only when $n \neq 0$, $(x \mapsto \sqrt{x})$ is a partial function on the reals with domain of definition $\{x \mid x \geq 0\}$, etc. A finite sequence $u \in A^*$ is a partial function $u : \mathbb{N} \rightarrow A$, as we defined it in 5.30. The empty set $\emptyset$ is (trivially) a partial function (with empty domain of definition!) and every (total) function on A to E is also a partial function, since (6-6) does not exclude $\text{Domain}(f) = A$,

$$\emptyset : A \rightarrow E, \quad f : A \rightarrow E \implies f : A \rightarrow E.$$

We will use systematically the convenient half-arrow notation for partial functions (recently established in computer science), as well as the common notations

$$f(x) \downarrow \iff_{\text{df}} x \in \text{Domain}(f), \quad f(x) \uparrow \iff_{\text{df}} x \notin \text{Domain}(f) \quad (6-7)$$

to indicate that a partial function is defined or undefined at some point; we sometimes read $f(x) \downarrow$ as $f(x)$ *converges* or f *converges at* x , and $f(x) \uparrow$ as $f(x)$ *diverges* or f *diverges at* x .

6.8. Definition. For each A and E ,

$$(A \rightarrow E) =_{\text{df}} \{f \subseteq A \times E \mid f : A \rightarrow E\} \quad (6-8)$$

is the set of all partial functions from A to E , in analogy with the notation $(A \rightarrow E)$ for the set of all (total) functions from A to E , cf. 4.14. The set $(A \rightarrow E)$ is partially ordered by the relation $\subseteq$,

$$f \subseteq g \iff (\forall x \in A)[f(x) \downarrow \implies [g(x) \downarrow \ \& \ f(x) = g(x)]],$$

with least element $\perp = \emptyset$.

6.9. Exercise. For each A, E ,

$$(A \rightarrow E) = \{f \upharpoonright X \mid f : A \rightarrow E \text{ \& } X \subseteq A\}.$$

Function restrictions are defined in (4-13).

It is harder to find least upper bounds for sets in these partial function posets than in powersets: if, for example, $S \subseteq (\mathbb{N} \rightarrow \mathbb{N})$ contains the two (total) constant functions $x \mapsto 0$ and $x \mapsto 1$, then any upper bound $f : \mathbb{N} \rightarrow \mathbb{N}$ of S would satisfy both $f(0) = 0$ and $f(0) = 1$, which is absurd. On the other hand, linearly ordered subsets of $(A \rightarrow E)$ have least upper bounds, and this is a fruitful property of these posets, worth a name.

6.10. Definition. A **chain** in a poset P is any linearly ordered subset S of P , i.e., a set satisfying

$$(\forall x, y \in S)[x \leq y \vee y \leq x].$$

A poset P is *chain-complete* or **inductive** if every chain in P has a least upper bound.

6.11. Exercise. The empty set is (trivially) a chain, hence every inductive poset P has a least element $\perp = \sup \emptyset$.

6.12. Exercise. Every flat poset is inductive; a discrete poset is inductive only when it has exactly one element, in which case it is also flat.

6.13. Exercise. The image $\{x_n \mid n \in \mathbb{N}\}$ of a non-decreasing sequence

$$x_0 \leq x_1 \leq x_2 \leq \cdots$$

is a chain; thus, every non-decreasing sequence has a **limit** in an inductive poset,

$$\lim_n x_n =_{\text{df}} \sup \{x_n \mid n \in \mathbb{N}\}. \quad (6-9)$$

6.14. Proposition. (1) For each set A , the powerset $\mathcal{P}(A)$ is inductive.

(2) For any two sets A, E , the poset $(A \rightarrow E)$ of all partial functions from A to E is inductive.

(3) For every poset P , the set

$$\text{Chains}(P) = \{S \subseteq P \mid S \text{ is a chain}\}$$

of all chains in P (partially ordered under $\subseteq$) is inductive.

PROOF. (1) is immediate by Exercise 6.6. (2) If $S \subseteq (A \rightarrow E)$ is a chain, then the union $\bigcup S$ is a partial function and obviously, $\bigcup S = \sup S$. (3) This is also proved by observing that the union of a chain of chains in a poset is also a chain. $\dashv$

6.15. Exercise. Neither $\mathbb{N}$ (with the usual ordering) nor the finite poset of Figure 6.2 are inductive.

6.16. Exercise. For each set E , the set $P = E^* \cup (\mathbb{N} \rightarrow E)$ of finite and infinite sequences from E is an inductive poset, under $\subseteq$.

6.17. Exercise. For any two sets A, E , the poset

$$(A \rightarrowtail E) =_{\text{df}} \{f \in (A \rightarrow E) \mid f \text{ is one-to-one}\}$$

of **partial injections** from A to E (partially ordered by $\subseteq$) is inductive.

We will find the most significant applications of the fixed point theorems in partial function posets; but the proofs will use only the fact that these posets are inductive, and there are lots of other interesting examples. Some of them are described in the problems at the end of the chapter.

Finally, we need to delineate the type of functions on inductive posets which must, necessarily, have fixed points.

6.18. Definition. A mapping¹⁵ $\pi : P \rightarrow Q$ on a poset P to another is **monotone** if for all $x, y \in P$,

$$x \leq_P y \implies \pi(x) \leq_Q \pi(y).$$

A monotone mapping need not be strictly increasing in the sense of

$$x <_P y \implies \pi(x) <_Q \pi(y),$$

e.g., every constant mapping is monotone.

Notice that if $\pi : P \rightarrow Q$ is monotone and $S \subseteq P$ is a chain, then the image $\pi[S]$ is also a chain; because given $x = \pi(u)$, $y = \pi(v)$ with $u, v \in S$, either $u \leq v$, which implies $x = \pi(u) \leq \pi(v) = y$, or $v \leq u$, which similarly implies $y \leq x$. This makes the next definition meaningful.

6.19. Definition. A monotone mapping $\pi : P \rightarrow Q$ on an inductive poset to another is **countably continuous** if for every non-empty, countable chain $S \subseteq P$,

$$\pi(\sup S) = \sup \pi[S].$$

6.20. Exercise. A monotone mapping $\pi : P \rightarrow Q$ on one inductive poset to another is countably continuous if and only if for every non-decreasing sequence $x_0 \leq_P x_1 \leq_P \dots$ of elements in P ,

$$\pi(\lim_n x_n) = \lim_n \pi(x_n).$$

Here the limit on the left is taken in P and the limit on the right is taken in Q .

6.21. Continuous Least Fixed Point Theorem. Every countably continuous, monotone mapping $\pi : P \rightarrow P$ on an inductive poset into itself has exactly one **strongly least fixed point** x^* , which is characterized by the two properties

$$\pi(x^*) = x^*, \tag{6-10}$$

$$(\forall y \in P)[\pi(y) \leq y \implies x^* \leq y]. \tag{6-11}$$

PROOF. The **orbit** of the least element $\perp$ under π is defined by the simple recursion on the natural numbers,

$$\begin{aligned} x_0 &= \perp, \\ x_{n+1} &= \pi(x_n). \end{aligned}$$

¹⁵It is convenient to refer to $\pi : P \rightarrow Q$ as a “mapping” rather than a “function” (which means the same thing), because in the interesting applications P is some partial function space ($A \rightarrow E$), Q may be another partial function space, π takes partial functions as arguments and (possibly) values and there are altogether too many functions around. Notice also that, pedantically, $\pi : \text{Field}(P) \rightarrow \text{Field}(Q)$ is a mapping from the field of P to that of Q .

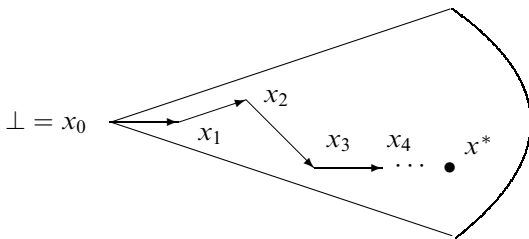


FIGURE 6.4. The Continuous Least Fixed Point Theorem.

Clearly $x_0 = \perp \leq x_1$, and by a trivial induction (using the monotonicity of π), for every n , $x_n \leq x_{n+1}$. Thus, the limit

$$x^* =_{\text{df}} \lim_n x_n = \sup \{x_n \mid n \in \mathbb{N}\} \quad (6-12)$$

exists by 6.13, and by the countable continuity of π ,

$$\pi(x^*) = \pi(\lim_n x_n) = \lim_n \pi(x_n) = \lim_n x_{n+1} = x^*.$$

For the second claimed property of x^* , we assume $\pi(y) \leq y$ and show by induction that for every n , $x_n \leq y$. **BASIS.** $x_0 = \perp \leq y$, trivially. **INDUCTION STEP.** The Induction Hypothesis gives us $x_n \leq y$, and we compute:

$$\begin{aligned} x_n \leq y &\implies \pi(x_n) \leq \pi(y), && \text{(because } \pi \text{ is monotone),} \\ &\implies x_{n+1} \leq \pi(y) \leq y, && \text{(by the assumption on } y). \end{aligned}$$

Thus, y is an upper bound of the set of values $\{x_n \mid n \in \mathbb{N}\}$ of the sequence, and hence, $x^* = \sup \{x_n \mid n \in \mathbb{N}\} \leq y$. $\dashv$

To apply the Continuous Least Fixed Point Theorem, we must formulate the problem at hand as a question of existence and (sometimes) uniqueness of solutions for an equation of the form $\pi(x) = x$, where $\pi : P \rightarrow P$ is monotone and countably continuous on some inductive poset P . This is typically the hardest part: to bring the problem in a form in which 6.21 can be applied. *Verification of the countable continuity of π is not necessary:* because we will show in the next chapter that 6.21 *remains true if we simply remove the hypothesis of countable continuity of π .* In any case, most applications involve simple monotone mappings on partial function posets for which it is often trivial to recognize a much stronger, natural continuity property.

6.22. Definition. A partial function $g : A \rightarrow E$ is **finite** if it has finite domain, i.e., if it is a finite set of ordered pairs.

A mapping $\pi : (A \rightarrow E) \rightarrow (B \rightarrow M)$ from one partial function space into another is **continuous**, if it is **monotone** and **compact**, i.e., for each $f : A \rightarrow E$, and all $y \in B$ and $v \in M$,

$$\pi(f)(y) = v \implies (\exists f_0 \subseteq f)[f_0 \text{ is finite \& } \pi(f_0)(y) = v]. \quad (6-13)$$

The notation is a bit convoluted, but what it means is quite simple: to compute $\pi(f)(y)$, we first compute the partial function $f' = \pi(f)$ and then we evaluate it at y , $\pi(f)(y) = f'(y)$: a monotone mapping π is continuous if each value $\pi(f)(y)$ of $\pi(f)$, whenever defined, depends only on finitely many values of f . In fact, we can combine the conditions of monotonicity and compactness in the following, simple characterization of continuity for partial function space mappings which often allows the immediate—“by inspection”—recognition that they are continuous.

6.23. Proposition. *A mapping $\pi : (A \rightarrow E) \rightarrow (B \rightarrow M)$ is continuous if and only if it satisfies both (6-13) and its converse, i.e., if for every $f : A \rightarrow E$ and all $y \in B$ and $v \in M$:*

$$\pi(f)(y) = v \iff (\exists f_0 \subseteq f)[f_0 \text{ is finite} \& \pi(f_0)(y) = v]. \quad (6-13^*)$$

For example, the mapping $\pi : (\mathbb{N} \rightarrow \mathbb{N}) \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$ defined by

$$\pi(f) = (n \mapsto f(n) + f(n^2))$$

is continuous, because (obviously) for every f and n ,

$$\pi(f)(n) = \pi(f_0)(n),$$

where f_0 is the restriction of f to the two-element set $\{n, n^2\}$.

PROOF OF 6.23. Suppose first that π is continuous, $f : A \rightarrow E$, $y \in B$, and $v \in M$. If $\pi(f)(y) = b$, then by the compactness of π , there exists some finite $f_0 \subseteq f$ such that $\pi(f_0)(y) = v$; and if such a finite $f_0 \subseteq f$ exists, then the monotonicity of π implies that $\pi(f)(y) = v$.

For the converse, assume that (6-13^{*}) holds for every $f : A \rightarrow E$ and all $y \in B$, $v \in M$. This gives immediately the compactness of π . To verify that π is monotone, suppose $f \subseteq g$ and $\pi(f)(y) = v$; by the $\implies$ direction of (6-13^{*}), there is some finite $f_0 \subseteq f$ such that $\pi(f_0)(y) = v$; but now $f_0 \subseteq g$, and so by the $\impliedby$ direction of (6-13^{*}), $\pi(g)(y) = v$. $\dashv$

6.24. Exercise. *Show that the mapping $\pi : (\mathbb{N} \rightarrow \mathbb{N}) \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$ defined by*

$$\pi(f) = (n \mapsto \sum_{i=0}^n f(i))$$

is continuous. Compute $\pi(n \mapsto 2n)(2)$ for this π .

6.25. Definition. A function $f : X \rightarrow Y$ from one topological space to another is (topologically) **continuous**, if the inverse image $f^{-1}[G]$ of every open subset of Y is an open subset of X . The definition of a **topological space** was sneaked in 4.30, as the first example of a structured set.

6.26. Exercise. *A function $f : X \rightarrow Y$ from one topological space to another is continuous if and only if the inverse image $f^{-1}[F]$ of every closed subset of Y is closed in X .*

One might guess that our using the term “continuous” in Definition 6.22 is not entirely accidental and that the notion of 6.22 has something to do with topological continuity. Indeed it does: the notions are equivalent when the

proper topology is put on partial function posets, but we will have no need of this fact and will leave it for Problem **x6.21**.

6.27. About topology. General (pointset) topology is to set theory like parsley to Greek food: some of it gets in almost every dish, but there are no great “parsley recipes” that the good Greek cook needs to know. Many notions and results of set theory are connected to topological ideas, but it is quite rare that you can prove an interesting theorem about sets by quoting some deep topological result. To avoid getting distracted with side issues, we will follow the general policy of giving the most direct, set theoretically natural definitions and proofs of the notions and results we need and leave the connections with topology for the problems. Occasionally the most natural approach is topological.

6.28. Lemma. *If $S \subseteq (A \rightarrow E)$ is a non-empty chain in a partial function poset and $f_0 \subseteq \sup S$ is a finite function, then there exists some $g \in S$ such that $f_0 \subseteq g$.*

PROOF is by induction on the number of elements in the domain of f_0 .

BASIS. $f_0 = \emptyset$ is the partial function which is nowhere defined. There is some $g \in S$ since S is non-empty, and $\emptyset \subseteq g$.

INDUCTION STEP. The domain of f_0 has $n + 1$ elements, so

$$f_0 = f_1 \cup \{(x, w)\} \subseteq \sup S,$$

where f_1 is a finite, partial function with just n elements in its domain, and by induction hypothesis, there exists some $g_1 \in S$ such that $f_1 \subseteq g_1$. Since $(x, w) \in \sup S$, there must also exist some $h' \in S$ such that $(x, w) \in h'$, and since S is a chain, either $g_1 \subseteq h'$ or $h' \subseteq g_1$; the g we need is the larger of these two partial functions. $\dashv$

6.29. Lemma. *Every continuous mapping $\pi : (A \rightarrow E) \rightarrow (B \rightarrow M)$ is countably continuous, in fact, for every (not necessarily countable) non-empty chain $S \subseteq (A \rightarrow E)$,*

$$\pi(\sup S) = \sup \pi[S].$$

PROOF. Suppose $S \subseteq (A \rightarrow E)$ is a non-empty chain with union $f = \sup S$. If $g \in S$, then $g \leq f$, and since π is monotone we have $\pi(g) \leq \pi(f)$, so that

$$\sup \pi[S] = \sup \{\pi(g) \mid g \in S\} \leq \pi(f).$$

For the converse inequality, we need to show that if $\pi(f)(y) = v$, then there exists some $g \in S$ such that $\pi(g)(y) = v$. By the continuity of π , there exists a finite $f_0 \subseteq f$, such that already $\pi(f_0)(y) = v$; by the preceding Lemma, there exists some $g \in S$ so that $f_0 \subseteq g$; and by the monotonicity of π , this implies $\pi(f_0) \subseteq \pi(g)$. In particular, since $\pi(f_0)(y) = v$, we have $\pi(g)(y) = v$, so this is the g we need. $\dashv$

The Continuous Least Fixed Point Theorem is evidently a simple corollary of the Recursion Theorem on the natural numbers **5.6**. In fact, it implies **5.6**

by a fairly direct argument, which is worth looking at, as it illustrates how we intend to apply **6.21**.

6.30. Proof of the Recursion Theorem from 6.21. For each given $a \in E$ and function $h : E \rightarrow E$, we define the mapping

$$\pi : (\mathbb{N} \rightarrow E) \rightarrow (\mathbb{N} \rightarrow E)$$

by the formula

$$\pi(f) = f', \text{ where } f'(x) = \begin{cases} a, & \text{if } x = 0, \\ h(f(x-1)), & \text{if } x > 0, \end{cases}$$

where f is any partial function from $\mathbb{N}$ to E and we understand the definition naturally, so that

$$x > 0 \implies [f'(x) \downarrow \iff h(f(x-1)) \downarrow \iff f(x-1) \downarrow].$$

Written out in detail, the mapping π associates a set of pairs $f' \subseteq (\mathbb{N} \times E)$ with every $f \in (\mathbb{N} \rightarrow E)$ and it is defined by the equation

$$\begin{aligned} \pi(f) = \{ & (0, a) \} \\ & \cup \{ (x, h(w)) \mid x > 0 \ \& \ (x-1, w) \in f \} \quad (f : \mathbb{N} \rightarrow E). \end{aligned} \quad (6-14)$$

From this we get that for every f and x ,

$$\pi(f)(x) = \pi(f_0)(x),$$

where $f_0 = \{(0, a)\}$ if $x = 0$ and $f_0 = \{(x-1, f(x-1))\}$ if $x > 0$, so that π is continuous by Proposition **6.23**, and hence countably continuous. Thus by **6.21**, it has a fixed point: that is, some partial function $f^* : \mathbb{N} \rightarrow E$ exists which satisfies $f^* = \pi(f^*)$, so that, immediately,

$$f^*(0) = a, \quad (6-15)$$

$$f^*(x+1) = h(f^*(x)) \quad (f^*(x) \downarrow). \quad (6-16)$$

Theorem **6.21** does not guarantee that this f^* is a *total function*, with domain of definition the entire $\mathbb{N}$, but this can be verified by an easy induction on x using the identities (6-15) and (6-16). $\dashv$

Consider next a case where it is not quite so obvious how to define the function we want directly by the Recursion Theorem.

6.31. Proposition. *For each function $h : \mathbb{N} \rightarrow \mathbb{N}$ and each infinite set $A \subseteq \mathbb{N}$ of natural numbers, there exists a (total) function $f : \mathbb{N} \rightarrow \mathbb{N}$ which satisfies the identity*

$$f(n) = \begin{cases} 0, & \text{if } n \in A, \\ h(f(n+1)), & \text{if } n \notin A. \end{cases} \quad (6-17)$$

PROOF. We define the mapping

$$\pi : (\mathbb{N} \rightarrow \mathbb{N}) \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$$

on the inductive poset of all unary, partial functions on $\mathbb{N}$ by the formula

$$\pi(f) = f', \text{ where } f'(n) = \begin{cases} 0, & \text{if } n \in A, \\ h(f(n+1)), & \text{if } n \notin A. \end{cases} \quad (6-18)$$

In full detail, this means we set

$$\pi(f) = \{(n, 0) \mid n \in A\} \cup \{(n, h(w)) \mid n \notin A \text{ \& } (n+1, w) \in f\},$$

which implies by inspection that π is continuous, hence countably continuous. Thus we must have a fixed point f which satisfies (6-17), and it is enough to prove that this f is total. Assume towards a contradiction that $f(n) \uparrow$ for some n . Notice that by (6-17), this means that $n \notin A$, else $f(n) \downarrow$, in fact, $f(n) = 0$. We will prove by induction on i that $f(n+i) \uparrow$, which implies again that for all i , $n+i \notin A$, so that $A \subseteq [0, n]$ is finite, contradicting the hypothesis. **BASES.** If $i = 0$, then $f(n+0) = f(n) \uparrow$, by assumption. **INDUCTION STEP.** Assume that $f(n+i) \uparrow$, so that by (6-17), once more, $n+i \notin A$. Now this implies that $f(n+i) = h(f(n+i+1))$ so that $f(n+i+1) \downarrow \implies f(n+1) \downarrow$ (since h is total), which violates the induction hypothesis. $\neg$

6.32. Exercise. Prove in detail that the mapping π in this proof is continuous.

As a third, typical application of the Continuous Least Fixed Point Theorem we consider the *Euclidean algorithm*.

6.33. Proposition. (1) There exists exactly one partial function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ with domain of definition $\{(n, m) \mid n, m \neq 0\}$ which satisfies the following identities for all $0 < n < m$:

$$\begin{aligned} f(m, n) &= f(n, m), \\ f(n, n) &= n, \\ f(n, m) &= f(n, m - n). \end{aligned} \quad (6-19)$$

(2) The unique f^* which satisfies the system (6-19) computes the greatest common divisor of any two natural numbers different from 0,

$$\begin{aligned} f^*(n, m) &= \gcd(n, m) \\ &= \text{the largest } k \text{ which divides evenly} \\ &\quad \text{both natural numbers } n, m. \end{aligned} \quad (6-20)$$

PROOF. With each partial function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ we associate the partial function $f' : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ which is defined by the formula

$$f'(n, m) = \begin{cases} f(m, n), & \text{if } n > m > 0, \\ n, & \text{if } n = m > 0, \\ f(n, m - n) & \text{if } 0 < n < m, \end{cases}$$

and we set

$$\pi(f) = f'.$$

The mapping $\pi : ((\mathbb{N} \times \mathbb{N}) \rightarrow \mathbb{N}) \rightarrow ((\mathbb{N} \times \mathbb{N}) \rightarrow \mathbb{N})$ is continuous, by inspection. It follows that there exists a least partial function $f^* : (\mathbb{N} \times \mathbb{N}) \rightarrow \mathbb{N}$ which satisfies

$$\pi(f^*) = f^*,$$

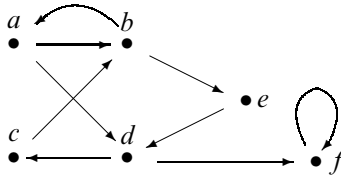


FIGURE 6.5

and this is (easily) equivalent with the system (6-19). Proof that for all $n, m \neq 0$

$$f^*(n, m) \downarrow \ \& \ f^*(n, m) = \gcd(n, m)$$

is by induction on the sum $n + m$. (Take cases whether $n > m > 0$, $n = m > 0$ or $0 < n < m$, and use the simple property of the natural numbers, that for $0 < n < m$, the common divisors of n, m are precisely the same as the common divisors of $n, m - n$.) $\dashv$

In this example we do not need the Least Fixed Point Theorem to prove the existence of a solution for the system (6-19), since we can verify directly that the function $\gcd$ is a solution. Despite this, the proposition is important because it yields a characterization of the function $\gcd$ which suggests a specific—and simple—method for computing it. For example, using only the identities of the system, we compute:

$$\begin{aligned} \gcd(231, 165) &= \gcd(165, 231) = \gcd(165, 66) = \gcd(66, 165) \\ &= \gcd(66, 99) = \gcd(66, 33) = \gcd(33, 66) \\ &= \gcd(33, 33) = 33. \end{aligned}$$

This computation of the value $\gcd(231, 165)$ is much simpler than the trivial one, where we would search for the greatest common divisor by testing in sequence all the numbers from 165 moving down, until we would find some common divisor of 165 and 231. The example is quite general: the characterization of a partial function f as the least solution of a system of simple identities typically yields an **algorithm**, a “recipe” for the “mechanical” computation of the values of f , and this is the underlying reason for the significance of the Continuous Least Fixed Point Theorem in theoretical computer science.

We end with a simple result about graphs which is related to the ideas of this chapter, see Problems **x6.16** and **x6.17**.

6.34. Definition. A **graph** is a structured set $(G, \rightarrow_G)$, where the set of **edges** $\rightarrow_G \subseteq G \times G$ is an arbitrary binary relation on the set of **nodes** G . The **transitive closure** of a graph G is the graph $\overline{G} = (G, \Rightarrow_G)$, where

$$\begin{aligned} x \Rightarrow_G y &\iff_{\text{df}} \text{there is a path from } x \text{ to } y \text{ in } G \\ &\iff (\exists z_0, \dots, z_n)[x = z_0 \rightarrow_G z_1 \ \& \ \dots \ \& \ z_{n-1} \rightarrow_G z_n = y]. \end{aligned}$$

We draw graphs much like posets, except we forget about the convention of “growing up or towards the right”: $x \rightarrow_G y$ holds if there is an arrow from x to y , and $x \Rightarrow_G y$ holds if you can move from x to y along the arrows of the diagram. In Figure 6.5 we have $f \rightarrow f$, $a \Rightarrow a$ and $a \Rightarrow c$, but $f \not\Rightarrow d$.

6.35. Proposition. *For each graph G , the transitive closure relation $\Rightarrow_G$ satisfies the equivalence*

$$x \Rightarrow_G y \iff x \rightarrow_G y \vee (\exists z \in G)[x \rightarrow_G z \ \& \ z \Rightarrow_G y]. \quad (6-21)$$

PROOF. Suppose first that (skipping the conjunction signs)

$$x = z_0 \rightarrow_G z_1 \rightarrow_G z_2 \rightarrow_G \cdots \rightarrow_G z_n = y;$$

if $n = 1$, we have $x \rightarrow_G y$, and if $n > 1$, then $x \rightarrow_G z_1$ and $z_1 \Rightarrow_G y$ (by the definition of $\Rightarrow_G$), so we have the right-hand side of (6-21), taking $z = z_1$. The converse is equally simple, taking cases on the two disjuncts of the right-hand side. $\dashv$

Problems for Chapter 6

x6.1. For every partial ordering $\leq$ on a set A , the **converse relation**

$$x \leq' y \iff_{\text{df}} y \leq x$$

is also a partial ordering. Of the inductive posets $(A \rightarrow E)$ and $\mathcal{P}(A)$, which one has an inductive, converse poset?

x6.2. Suppose $\leq_E$ is an inductive partial ordering on the set E , A is a set and $\leq$ is the “pointwise” partial ordering on the function space $(A \rightarrow E)$,

$$f \leq g \iff_{\text{df}} (\forall x \in A)[f(x) \leq_E g(x)] \quad (f, g : A \rightarrow E).$$

Prove that $\leq$ is an inductive partial ordering on $(A \rightarrow E)$.

x6.3. If the partial orderings $\leq_1, \leq_2$ on the respective sets P_1, P_2 are inductive, then the following relation $\leq$ on the Cartesian product $P_1 \times P_2$ is also inductive:

$$(x_1, x_2) \leq (y_1, y_2) \iff_{\text{df}} x_1 \leq_1 y_1 \ \& \ x_2 \leq_2 y_2.$$

With this partial ordering, the poset $P_1 \times P_2$ is called **the product** of the two posets P_1 and P_2 .

x6.4. Suppose P_1, P_2, Q are inductive posets. A mapping $\pi : P_1 \times P_2 \rightarrow Q$ is **separately monotone** if for each $x_1 \in P_1$, the mapping $(x_2 \mapsto \pi(x_1, x_2))$ on P_2 is monotone, and symmetrically for each $x_2 \in P_2$. Prove that π is monotone (on the product poset) if and only if it is separately monotone,

x6.5. Suppose P_1, P_2, Q are inductive posets. A mapping $\pi : P_1 \times P_2 \rightarrow Q$ is **separately countably continuous**, if for each $x_1 \in P_1$, the mapping $(x_2 \mapsto \pi(x_1, x_2))$ on P_2 is countably continuous, and symmetrically for each $x_2 \in P_2$. Prove that π is countably continuous if and only if it is separately countably continuous.

6.36. Definition. A point M is **maximal** in a subset S of a poset P if M is a member of S and no member of S is bigger,

$$M \in S \ \& \ (\forall x \in S)[M \leq x \implies M = x].$$

A point m is **minimal** in S if it is a member of S and no member of S is smaller,

$$m \in S \ \& \ (\forall x \in S)[x \leq m \implies x = m].$$

x6.6. Find in the poset of Figure 6.2 a subset S which has a maximal element but no maximum and another subset S' which has a minimal element but no minimum.

***x6.7.** Every finite, non-empty subset of an arbitrary poset P has at least one maximal and one minimal member.

***x6.8.** A finite poset P is inductive if and only if it has a least element.

An important notion in computer science is that of a *stream*, for example the stream of bytes in a file transmitted over the telephone lines to my home computer from the University of Athens CYBER. A stream is basically a sequence, but it may be *infinite*, in the idealized case; *terminated*, if after some stage an end-of-file signal comes and my machine knows that the transmission is done; or *stalled*, if after some stage the bytes stop coming, without warning, perhaps because the CYBER died or the telephone connection was interrupted.¹⁶

6.37. Definition. For each set A , we fix some $t \notin A$ (for example, the object $r(A)$ of (3-4)) and we define the **streams from A** by:

$\text{Streams}(A)$

$$=_{\text{df}} \{ \sigma : \mathbb{N} \rightarrow A \cup \{t\} \mid (\forall i < j)[\sigma(j) \downarrow \implies [\sigma(i) \downarrow \ \& \ \sigma(i) \neq t]] \}.$$

We call a stream σ **terminated** or **convergent** if for some n , $\sigma(n) = t$, in which case, by the definition $\text{Domain}(\sigma) = [0, n + 1)$; **infinite** if $\text{Domain}(\sigma) = \mathbb{N}$; and **stalled** if $\text{Domain}(\sigma)$ is a finite, initial segment of $\mathbb{N}$ but σ does not take on the *terminating value* t . The infinite and stalled streams together are called **divergent**.

x6.9. For each set A , the set of streams $\text{Streams}(A)$ is an inductive poset under the natural, partial ordering $\sqsubseteq$, where, as for strings,

$$\sigma \sqsubseteq \tau \iff_{\text{df}} \sigma \subseteq \tau. \tag{6-22}$$

What are its maximal elements?

x6.10. The **concatenation** $\sigma \star \tau$ of two streams is defined so that if σ is divergent, then $\sigma \star \tau = \sigma$ and if σ is convergent with domain $[0, n + 1)$, then

$$i < n \implies (\sigma \star \tau)(i) = \sigma(i), \quad (\sigma \star \tau)(n + i) = \tau(i).$$

¹⁶Well, the CYBER has died completely since the first edition of these Notes, but the newer machines and more robust telephone lines still quit unexpectedly on some occasions!

Prove that $\star$ is a continuous function (of two variables) on $\text{Streams}(A)$.

The full Least Fixed Point Theorem can be proved directly and easily for powersets:

***x6.11.** Suppose $\pi : \mathcal{P}(A) \rightarrow \mathcal{P}(A)$ is a monotone mapping on a powerset. Prove that the set

$$A^\star = \bigcap \{X \mid \pi(X) \subseteq X\}$$

is the least fixed point of π , and

$$A_\star = \bigcup \{X \mid X \subseteq \pi(X)\}$$

is the largest fixed point of π .

The next few problems deal with “algorithmic” applications of the Least Fixed Point Theorem.

x6.12. For each relation $R \subseteq \mathbb{N} \times A$, there exists a least partial function $f : \mathbb{N} \times A \rightarrow \mathbb{N}$ such that

$$\begin{cases} R(n, x) \implies f(n, x) = n, \\ \neg R(n, x) \implies f(n, x) = f(n+1, x), \end{cases}$$

It follows that

$$\begin{aligned} f(n, x) \downarrow &\iff (\exists m \geq n)[R(m, x), \\ f(n, x) \downarrow &\implies f(n, x) = \text{the least } m \geq n \text{ such that } R(m, x). \end{aligned}$$

x6.13. For any three partial functions f_0, g, h with domains and ranges such that the identities below make sense, there exists a least partial function $f : \mathbb{N} \times A \rightarrow E$ which satisfies the identities

$$\begin{aligned} f(0, x) &= f_0(x), \\ f(n+1, x) &= h(f(n, g(n, x)), n, x). \end{aligned}$$

x6.14. Prove that there is exactly one total function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ which satisfies the identities

$$\begin{aligned} f(0, n) &= f(n, 0) = 0, \\ f(n+1, m+1) &= f(n, m) + 1. \end{aligned}$$

Compute $f(5, 23)$ using these identities and “explain” what $f(n, m)$ is, for any n, m .

6.38. Definition. On the set E^* of strings (finite sequences) from a set E defined in 5.30, we define the partial functions

$$\text{head}(u) = u(0), \tag{6-23}$$

$$\text{tail}(u) = \langle u(1), \dots, u(\text{lh}(u) - 1) \rangle. \tag{6-24}$$

Notice that $\text{head}(u) \downarrow$ when $\text{lh}(u) > 0$, while $\text{tail}(u)$ is always defined, but it is the empty string when $\text{lh}(u) \leq 1$.

x6.15. Prove that there exists a unique, total function $r : E^* \rightarrow E^*$ which satisfies the identity

$$r(u) = \begin{cases} u, & \text{if } \text{lh}(u) \leq 1, \\ r(\text{tail}(u)) \star \langle \text{head}(u) \rangle, & \text{if } \text{lh}(u) > 1. \end{cases}$$

Compute $r(\langle a, b, c \rangle)$ and describe $r(u)$ in general.

x6.16. Prove that for each graph G , the relation $\Rightarrow_G$ is the least (under $\subseteq$) transitive relation on G which includes the edge relation $\rightarrow_G$.

x6.17. Prove that for every graph G with edge relation $\rightarrow_G$, the relation $\Rightarrow_G$ is the common least fixed point of the following monotone operators on the poset $\mathcal{P}(G \times G)$ of all binary relations on G :

$$\begin{aligned} \pi_1(R) &= \{(x, y) \mid x \rightarrow_G y \vee (\exists z)[x \rightarrow_G z \ \& \ (z, y) \in R]\}, \\ \pi_2(R) &= \{(x, y) \mid x \rightarrow_G y \vee (\exists z)[(x, z) \in R \ \& \ z \rightarrow_G y]\}, \\ \pi_3(R) &= \{(x, y) \mid x \rightarrow_G y \vee (\exists z)[x \rightarrow_G z \rightarrow_G y] \\ &\quad \vee (\exists z, w)[x \rightarrow_G z \ \& \ (z, w) \in R \ \& \ w \rightarrow_G y]\}. \end{aligned}$$

x6.18. Let P_1, P_2 be inductive posets and

$$\begin{aligned} \pi_1 : P_1 \times P_2 &\rightarrow P_1, \\ \pi_2 : P_1 \times P_2 &\rightarrow P_2 \end{aligned}$$

arbitrary countably continuous, monotone mappings, where $P_1 \times P_2$ is the product. Prove that there exist unique **least, mutual or simultaneous fixed points** x_1^*, x_2^* which are characterized by the properties:

$$\begin{aligned} \pi_1(x_1^*, x_2^*) &= x_1^*, \quad \pi_2(x_1^*, x_2^*) = x_2^*, \\ \pi_1(y_1, y_2) \leq_1 y_1 \ \& \ \pi_2(y_1, y_2) \leq_2 y_2 &\implies x_1^* \leq_1 y_1 \ \& \ x_2^* \leq_2 y_2. \end{aligned}$$

The next problem is an algorithmic version of the well-known number theoretic result, that for any two natural numbers $n, m \neq 0$, there exist (positive or negative) integers α, β such that

$$\gcd(n, m) = \alpha n + \beta m.$$

The proof uses some simple properties of the set of rational integers

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

***x6.19.** There exists exactly one pair of partial functions

$$\alpha : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{Z}, \quad \beta : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{Z},$$

with common domain of definition $\{(n, m) \mid n, m \neq 0\}$ which satisfy the following identities for all $n, m, k > 0$:

$$\begin{aligned} \text{if } n \neq m, \text{ then } \alpha(n, m) &= \beta(m, n), \\ \alpha(n, n) &= 1, \quad \beta(n, n) = 0, \\ \alpha(n, n+k) &= \alpha(n, k) - \beta(n, k), \quad \beta(n, n+k) = \beta(n, k). \end{aligned}$$

It follows that for any two natural numbers $n, m \neq 0$,

$$\gcd(n, m) = \alpha(n, m)n + \beta(n, m)m.$$

x6.20. Find integers $\alpha, \beta \in \mathbb{Z}$ such that

$$33 = 231\alpha + 165\beta.$$

Find also integers $\alpha_1, \beta_1 \in \mathbb{Z}$ such that

$$1 = 137\alpha_1 + 997\beta_1.$$

6.39. Definition. For each finite partial function $g : A \rightarrow E$, the **neighborhood** determined by g in the poset $(A \rightarrow E)$ is the set

$$N(g) =_{\text{df}} \{f : A \rightarrow E \mid g \subseteq f\}$$

of all extensions of g . A set $G \subseteq (A \rightarrow E)$ is **open in the topology of pointwise convergence** if

$$f \in G \implies (\exists g, \text{finite})[f \in N(g) \subseteq G].$$

x6.21. Prove that the family of open sets in $(A \rightarrow E)$ defined in **6.39** is a topology by **4.30**, and a mapping

$$\pi : (A \rightarrow E) \rightarrow (B \rightarrow M)$$

is continuous in this topology by **6.25** if and only if it is continuous by **6.22**.

6.40. Definition. A subset $G \subseteq P$ of an inductive poset is **Scott open** if (1) it is upward closed, i.e.,

$$x \in G \ \& \ x \leq y \implies y \in G,$$

and (2) for every non-empty chain $S \subseteq P$,

$$\sup S \in G \implies (\exists x \in S)[x \in G].$$

***x6.22.** Prove that the family of Scott open subsets of an inductive poset P is a topology.

***x6.23.** Suppose P and Q are inductive posets and $\pi : P \rightarrow Q$ is a mapping. Prove that π is continuous in the relevant Scott topologies if and only if π is monotone and for every non-empty chain $S \subseteq P$,

$$\pi(\sup S) = \sup \pi[S].$$

HINT. Some find it helpful for this problem to first prove and use the fact that for every $c \in P$, the set $\{x \in P \mid x \leq c\}$ is Scott closed.

***x6.24.** Suppose A is a countable set and $\pi : (A \rightarrow E) \rightarrow (B \rightarrow M)$ is a mapping. Show that π is continuous (by the definition in **6.22**) if and only if it is continuous with respect to the Scott topologies in the posets $(A \rightarrow E)$ and $(B \rightarrow M)$.

The Continuous Least Fixed Point Theorem is often formulated for the class of *directed-complete* posets, particularly in Computer Science texts.

6.41. Definition. A subset $S \subseteq P$ of a poset P is **directed** if any two members of S have an upper bound in S ,

$$x, y \in S \implies (\exists z \in S)[x \leq z \ \& \ y \leq z].$$

A poset P is **directed-complete** (a **dcpo**) if every directed $S \subseteq P$ has a least upper bound.

x6.25. Every chain in a poset is a directed set, hence, every dcpo is an inductive poset and the least fixed point theorems hold for directed-complete posets.

x6.26. For each A and E , the posets $(A \rightarrow E)$ and $(A \multimap E)$ are directed-complete.

x6.27. A mapping $\pi : (A \rightarrow E) \rightarrow (B \rightarrow M)$ is continuous if and only if for each directed $S \subseteq (A \rightarrow E)$,

$$\pi(\sup S) = \sup \pi[S].$$

x6.28. The product $P_1 \times P_2$ (Problem **x6.3**) of two directed-complete partial orderings is also directed-complete.

***x6.29.** Every countable, inductive poset is directed-complete.

Actually the notions of *inductive* and *directed-complete* are equivalent; for a monotone mapping $\pi : P \rightarrow Q$ on one inductive poset to another, the equation

$$\pi(\sup S) = \sup \pi[S] \tag{6-25}$$

holds for all non-empty chains $S \subseteq P$ if and only if it holds for all non-empty directed sets $S \subseteq P$; and the characterization of Scott continuity in **x6.24** holds whether A is countable or not. The proofs of these results are not elementary and require the Axiom of Choice, see Problems **x9.22** – **x9.25**.